

Formal Modeling

Based on the Rigorous Software Engineering Exercises @ ETH Zürich

Task 1 (Modeling a University). Consider the university system described below:

- There are undergraduate and graduate students. No student is both an undergraduate and a graduate student.
 - A student should register at a university. Registered students are considered legal, while unregistered students are not.
 - Every student has a unique student ID, and he or she has exactly one major subject. There must not be any unassigned student ID.
 - Students registered at the same university with the same major are classmates.
 - Graduates and undergraduates are never classmates. Also, students from different universities or with different majors are never classmates.
 - A student cannot be his/her own classmate.
- a) Think about how to structure this system in terms of classes and relations (i.e., pointers between classes). Draw a class diagram. The fact whether a student is legal should be encoded using a simple boolean flag.
- b) Create an Alloy model of the university system, reflecting your structure from subtask (a). Visualize the model for 2 universities, 3 majors, 3 students, and 3 IDs by running Alloy.¹

Task 2 (UZH Bus). Consider the following Alloy model of a UZH shuttle bus that connects the university campuses. The model is incorrect.

```
1  sig UZHBusStation {
2    next: set UZHBusStation
3  }
4
5  one sig ZurichCity, Irchel, Oerlikon, Schlieren in UZHBusStation {}
```

¹Instructions how to download and run Alloy can be found here: <https://alloytools.org/download.html>

```

6
7 sig UZHBus {
8   station: lone UZHBusStation
9 }
10
11 fact {
12   no (Oerlikon - Schlieren)
13   all s: UZHBusStation | UZHBusStation in s.^next and some s.next
14   all b1, b2: UZHBus | b1.station != b2.station
15 }
16
17 pred show {}

```

- a) Without running the Alloy analyzer, find and visualize an instance produced by the command `run show for 2 but 1 UZHBusStation`.
- b) Consider the following list of specifications (i–v), not all of which are currently fulfilled by the given Alloy model. Fix the Alloy model such that all specifications are fulfilled. You are allowed to use the Alloy analyzer.
 - (i) There are exactly 4 UZHBusStations: ZurichCity, Irchel, Oerlikon, and Schlieren.
 - (ii) The UZHBusStations form one directed circle.
 - (iii) There is no UZHBusStation between Oerlikon and Schlieren.
 - (iv) There can be at most one UZHBus at each UZHBusStation.
 - (v) There is at least one UZHBus.

Hint: Only a few local changes to the model are required.